MIAMBILI DUCK — Network Infrastructure Overview

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1. Purpose & Scope

This document describes the network infrastructure for MIAMBILI DUCK, a retail chain operating a head office, five retail stores, and one warehouse, all located within Nairobi. It covers the site inventory, wide-area connectivity model, device types in use, and availability targets for each location.

2. Executive Summary

MIAMBILI DUCK's network is built on a hub-and-spoke SD-WAN architecture. The head office (HQ) in Nairobi hosts the SD-WAN hub and sits behind a dedicated HQ firewall, with a cloud overlay extending connectivity to all branch sites. Each of the five retail stores and the warehouse connects back to the HQ hub over an SD-WAN/IPsec VPN tunnel, rather than through individual site-to-site links. All seven sites carry a uniform 99% uptime SLA target.

3. Site Inventory

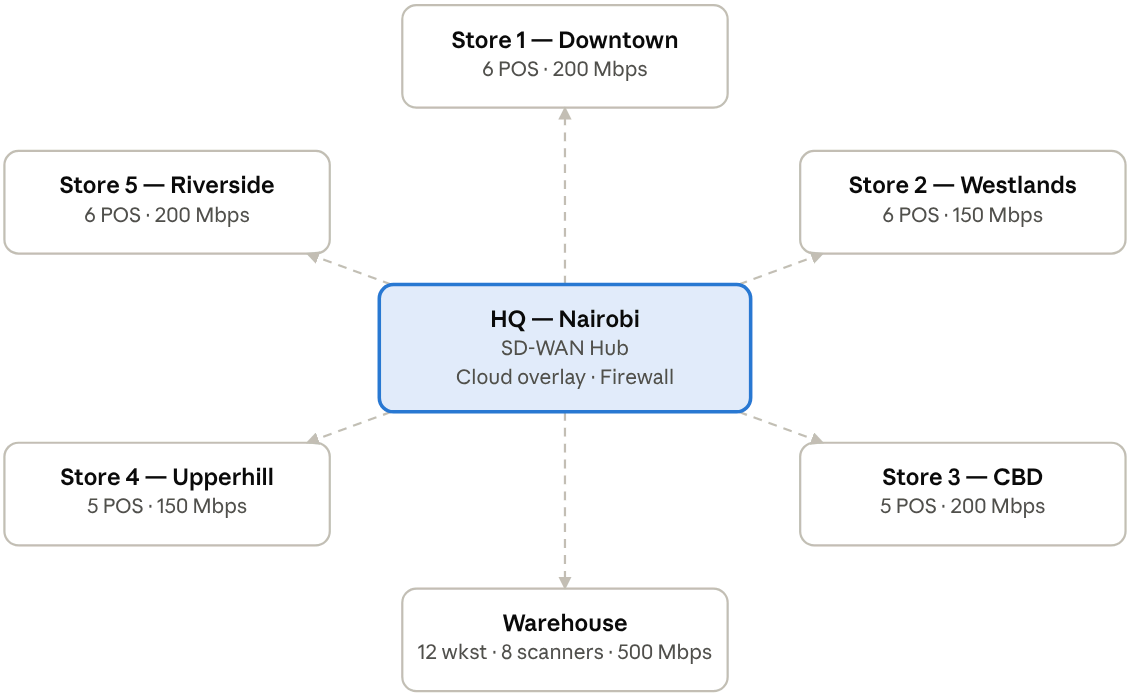
Site	Role	Endpoints	Bandwidth	Departments / Function	SLA Target
HQ — Nairobi	Hub site	25 workstations, 2 servers	500 Mbps (fiber)	Admin, Finance, IT	99%
Store 1 — Downtown	Retail	6 POS terminals, 4 back-office PCs	200 Mbps (business fiber)	Retail	99%
Store 2 — Westlands	Retail	6 POS terminals, 3 back-office PCs	150 Mbps (business fiber)	Retail	99%
Store 3 — CBD	Retail	5 POS terminals, 3 back-office PCs	200 Mbps (business fiber)	Retail	99%
Store 4 — Upperhill	Retail	5 POS terminals, 3 back-office PCs	150 Mbps (business fiber)	Retail	99%

Site	Role	Endpoints	Bandwidth	Departments / Function	SLA Target
Store 5 — Riverside	Retail	6 POS terminals, 3 back-office PCs	200 Mbps (business fiber)	Retail	99%
Warehouse	Logistics	12 workstations, 8 handheld scanners	500 Mbps (fiber)	Inventory, fulfillment	99%

4. Network Topology

Six branch sites connect to HQ through a hub-and-spoke SD-WAN overlay

Dashed lines = SD-WAN / IPsec VPN overlay · 99% uptime SLA at every site



hub-and-spoke SD-WAN topology · HQ + 6 sites

HQ sits at the centre of the overlay; every branch site reaches it over its own SD-WAN/IPsec VPN tunnel.

5. WAN Connectivity Model

- **Topology:** Hub-and-spoke, with the SD-WAN hub located at HQ and extended through a cloud overlay.

- **Site links:** Every branch site (5 stores + warehouse) connects to the HQ/cloud overlay via an SD-WAN/IPsec VPN tunnel. There are no direct site-to-site links; all inter-site traffic routes through the hub.
- **Perimeter:** HQ sits behind a dedicated firewall, which is the entry/exit point for the SD-WAN hub and the cloud overlay.
- **Availability:** A 99% uptime SLA applies uniformly across all seven sites.

6. Endpoint & Device Types

The following device classes are present across the network:

- PCs (workstations and back-office terminals)
- Servers (HQ only)
- Network switches
- Wireless access points
- VoIP phones
- Smartphones (mobile/BYOD)
- Printers
- Laptops
- POS terminals and handheld scanners (retail and warehouse sites)

7. Segmentation & Addressing — Recommendations

Per-site VLAN segmentation and IP addressing were part of the design intent but are not fully captured in the source material for this document. As a baseline recommendation:

- Separate VLANs per function at each site (e.g., POS/retail traffic isolated from back-office and guest/Wi-Fi traffic).
- A consistent, non-overlapping RFC1918 addressing scheme per site to simplify the SD-WAN overlay's routing and avoid conflicts if sites are ever bridged directly.
- Management/IT traffic at HQ kept on its own VLAN, isolated from retail POS traffic for PCI-related segmentation.

(The source diagram references VLAN and IP addressing labels; specific allocation values were not legible in the source and should be inserted here once available.)

8. Notes & Assumptions

- All bandwidth figures reflect each site's primary fiber/business fiber circuit.
- The 99% SLA target is stated uniformly across all sites in the source material; no differentiated tiers were specified.
- This document reflects the infrastructure overview as provided; it does not include firewall rule sets, routing protocol configuration, or physical cabling detail.